

Yasir Hassan

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EDUCATION

University of Texas at Austin

Bachelor of Science in Electrical and Computer Engineering

Austin, TX

May 2027

Relevant Coursework: Data Structures, Algorithms, Digital Logic Design, Discrete Mathematics, Probability, Computer Architecture

EXPERIENCE

Spectrum

Software Engineer Intern

Austin, TX

May 2025 – Aug 2025

- Contributed to a network traffic anomaly detection effort, helping build and evaluate models that flagged unusual patterns in log data for the diagnostics team.
- Worked on conversational AI and agent assist tooling that helped support representatives resolve customer issues faster, contributing to a 54% reduction in average resolution time.

Aristocrat Leisure

Software Engineer Intern

Austin, TX

Jan 2024 – May 2024

- Built a GPU usage monitoring tool in C++ and OpenGL that surfaced rendering performance across different hardware configurations, giving users visibility into system behavior.
- Collaborated with the engineering team using Git and GitHub to debug and refine the tool.

Preptify

Software Engineer Intern

Austin, TX

Sep 2023 – Dec 2023

- Built the frontend of an early stage mock interview platform for software and electrical engineering candidates using HTML, CSS, and JavaScript, contributing to a 25% increase in users trying the product after launch.

PROJECTS

MarketPulse: Full Stack Financial ML Platform (Python, FastAPI, React, TypeScript, scikit-learn, Docker)

- Built a full stack platform combining near real time market data, NLP sentiment analysis (VADER/FinBERT), and explainable ML stock predictions with a React/TypeScript frontend and FastAPI backend, verified by 96 backend tests at 89% coverage plus Vitest and Playwright suites.
- Designed a leakage free feature pipeline with walk-forward validation and a backtesting engine that models transaction costs and slippage against buy and hold and baseline strategies.
- Diagnosed and fixed a probability calibration bug where the model reported 97% confidence on near coin flip accuracy, then added a regression test to prevent recurrence.

Marginalia: Retrieval Augmented Generation Assistant (Python, FastAPI, ChromaDB, sentence-transformers, SQLite)

- Built a RAG assistant that answers natural language questions over personal notes and PDFs with citations grounded in source documents, streaming answers over SSE with pluggable LLM backends (Anthropic, Ollama).
- Implemented incremental indexing using a SQLite manifest (mtime/size checks with SHA-256 fallback) and a debounced folder watcher so unchanged files are never re-embedded, and added a precision@k / recall@k retrieval evaluation harness.

Real Time News Classifier (Python, scikit-learn, AWS, FastAPI)

- Built a topic classification pipeline (TF-IDF + Logistic Regression) for news articles, deployed on AWS (S3, Lambda, EC2) with automated retraining via GitHub Actions.

LEADERSHIP

Gamma Beta Fraternity

Vice President of Service

Austin, TX

Jan 2024 – Present

- Planned and led community service initiatives, coordinating logistics, outreach, and post event feedback to improve future programming.
- Built a semester long service roadmap and used member input to prioritize high impact events and grow participation.

SKILLS

Languages: Python, C++, TypeScript, JavaScript, SQL, Verilog

Frameworks/Libraries: React, FastAPI, Flask, scikit-learn, Pandas, NumPy, PyTorch, SQLAlchemy, OpenGL

Tools: Git, GitHub Actions, Docker, Linux, Jira, Pytest, Playwright